TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC5701 - Compilers & Interpreters

Regular Expressions

Dr. Francisco J. Torres-Rojas

Javier Lee Liang
2020184824

Steven D. Loaíciga Segura
2024214083

Luan Valverde Villalobos
2024080542

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Exercises to solve

Page of the exercises

12. The set of strings over $\{a, b, c\}$ in which all the a's precede the b's, which in turn precede the c's. It is possible that there are no a's, b's, or c's.
13. The same set as Exercise 12 without the null string.
14. The set of strings of length two or more over $\{a, b\}$ in which all the a's precede the b's.
15. The set of strings over $\{a, b\}$ that contain the substring aa and the substring bb.
16. The set of strings over $\{a, b\}$ in which the substring aa occurs at least twice. *Hint:* Beware of the substring aaa.
17. The set of strings over $\{a, b, c\}$ that do not contain the substring aa.
18. The set of strings over $\{a, b\}$ that do not begin with the substring aaa.
19. The set of strings over $\{a, b\}$ that do not contain the substring aaa.
20. The set of strings over $\{a, b\}$ that do not contain the substring aba.
21. The set of strings over $\{a, b\}$ in which the substring aa occurs exactly once.
22. The set of strings over $\{a, b, c\}$ that begin with a, contain exactly two b's, and end with cc.
23. The set of strings over $\{a, b\}$ that contain the substring ab and the substring ba.
24. The set of strings over $\{a, b, c\}$ that contain the substrings aa, bb, and cc.
25. The set of strings over $\{a, b, c\}$ in which every b is immediately followed by at least one c.
26. The set of strings over $\{a, b, c\}$ with length three.
27. The set of strings over $\{a, b, c\}$ with length less than three.
28. The set of strings over $\{a, b, c\}$ with length greater than three.
29. The set of strings over $\{a, b\}$ in which the number of a's is divisible by three.
30. The set of strings over $\{a, b, c\}$ in which the total number of b's and c's is three.
31. The set of strings over $\{a, b\}$ in which every a is either immediately preceded or immediately followed by b, for example, $baab$, aba, and b.
32. The set of strings of odd length over $\{a, b\}$ that contain the substring bb.
33. The set of strings of even length over $\{a, b, c\}$ that contain exactly one a.
34. The set of strings over $\{a, b, c\}$ with an odd number of occurrences of the substring ab.
35. The set of strings of odd length over $\{a, b\}$ that contain exactly two b's.
36. The set of strings over $\{a, b\}$ with an even number of a's or an odd number of b's.

Figure 1: Exercises to solve. Starting from the number 12 to 36

Exercises solved

Exercise 12

The set of strings over $\{a,b,c\}$ in which all the a's precede the b's, which in turn precede the c's. It is possible that there are no a's, b's, or c's.

$$a^*b^*c^*$$

Exercise 13

The same set as Exercise 12 without the null string.

$$aa^*b^*c^* \mid a^*bb^*c^* \mid a^*b^*cc^*$$

Exercise 14

The set of strings of length two or more over $\{a,b\}$ in which all the a's precede the b's.

$$(aaa^*b^*) \mid (aa^*bb^*) \mid (bbb^*)$$

Exercise 15

The set of strings over $\{a,b\}$ that contain the substring aa and the substring bb.

$$(a \mid b)^*aa(a \mid b)^*bb(a \mid b)^* \mid (a \mid b)^*bb(a \mid b)^*aa(a \mid b)^*$$

Exercise 16

The set of strings over $\{a,b\}$ in which the substring aa occurs at least twice. *Hint*: Beware of the substring aaa.

$$b^*aab^*aab^*$$

Exercise 17

The set of strings over $\{a,b,c\}$ that do not contain the substring aa.

$$(b \mid c \mid ab \mid ac)^* \mid (b \mid c \mid ab \mid ac)^*a$$

Exercise 18

The set of strings over $\{a,b\}$ that do not begin with the substring aaa.

$$(a \mid b \mid c) \mid (a \mid b \mid c)(a \mid b \mid c) \mid (b \mid c)(a \mid b \mid c)^* \mid a(b \mid c)(a \mid b \mid c)^* \mid aa(b \mid c)(a \mid b \mid c)^*$$

Exercise 19

The set of strings over $\{a,b\}$ that do not contain the substring aaa.

$$(b^* \mid ab \mid aab)^* \mid (b^* \mid ab \mid aab)^*(a \mid aa)$$

Exercise 20

The set of strings over $\{a,b\}$ that do not contain the substring aba.

$$(b \mid a^*)(bbb^* \mid a^*)^* \mid (b \mid a^*)(bbb^* \mid a^*)^*b$$

Exercise 21

The set of strings over $\{a,b\}$ in which the substring aa occurs extacly once.

$$(b \mid ab)^*aa(b \mid ba)^*$$

Exercise 22

The set of strings over $\{a,b,c\}$ that begin with a, contain exactly two b's, and end with cc.

$$a(a \mid c)^*b(a \mid c)^*b(a \mid c)^*cc$$

Exercise 23

The set of strings over $\{a,b\}$ that contain the substring ab and the substring ba.

$$(a \mid b)^*(ab(a \mid b)^*ba \mid ba(a \mid b)^*ab \mid aba \mid bab)(a \mid b)^*$$

Exercise 24

The set of strings over $\{a,b,c\}$ that contain the substrings aa, bb and cc.

$$\begin{aligned}(a \mid b \mid c)^*\Big(&aa(a \mid b \mid c)^*bb(a \mid b \mid c)^*cc\\ &\mid aa(a \mid b \mid c)^*cc(a \mid b \mid c)^*bb\\ &\mid bb(a \mid b \mid c)^*aa(a \mid b \mid c)^*cc\\ &\mid bb(a \mid b \mid c)^*cc(a \mid b \mid c)^*aa\\ &\mid cc(a \mid b \mid c)^*aa(a \mid b \mid c)^*bb\\ &\mid cc(a \mid b \mid c)^*bb(a \mid b \mid c)^*aa\Big)(a \mid b \mid c)^*\end{aligned}$$

Exercise 25

The set of strings over $\{a,b,c\}$ in which every b is immediately followed by at least one c.

$$(a \mid c \mid bc)^*$$

Exercise 26

The set of strings over $\{a,b,c\}$ with length three.

$$(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)$$

Exercise 27

The set of strings over $\{a,b,c\}$ with length less than three.

$$(a \mid b \mid c) \mid (a \mid b \mid c)(a \mid b \mid c) \mid \varepsilon$$

Exercise 28

The set of strings over $\{a,b,c\}$ with length greater than three.

$$(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)^*$$

Exercise 29

The set of strings over $\{a,b\}$ in which the number of a's is divisible by three.

$$(b^*ab^*ab^*a)^*$$

Exercise 30

The set of strings over $\{a,b,c\}$ in which the total number of a's is divisible by three.

$$\varepsilon \mid a^*(b \mid c)a^*(b \mid c)a^*(b \mid c)a^*$$

Exercise 31

The set of strings over $\{a,b\}$ in which every a is either immediately preceded or immediately followed by b, for example, $baab$, aba, and b.

$$\varepsilon \mid (b \mid ab \mid ba)^*$$

Exercise 32

The set of strings of odd length over $\{a,b\}$ that contain the substring bb.

$$((a \mid b)(a \mid b))^*bb((a \mid b)(a \mid b))^*(a \mid b) \mid ((a \mid b)(a \mid b))^*(a \mid b)bb((a \mid b)(a \mid b))^*$$

Exercise 33

The set of strings of even length over $\{a,b,c\}$ that contain exactly one a.

$$((b \mid c)(b \mid c))^*a(((b \mid c)(b \mid c))^*(b \mid c)) \mid ((b \mid c)(b \mid c))^*(b \mid c)a((b \mid c)(b \mid c))^*$$

Exercise 34

The set of strings over $\{a,b,c\}$ with an odd number of occurrences of the substring ab.

$$((b \mid a^*c)^*a^*)ab((b \mid a^*c)^*a^*)(ab((b \mid a^*c)^*a^*)ab((b \mid a^*c)^*a^*))^*$$

Exercise 35

The set of strings of odd length over $\{a,b\}$ that contain exactly two b's.

$(aa)^*ab(aa)^*b(aa)^* \mid (aa)^*b(aa)^*ab(aa) \mid (aa)^*b(aa)^*b(aa)^*a \mid (aa)^*ab(aa)^*ab(aa)^*a$

Exercise 36

The set of strings over $\{a,b\}$ with an even number of a's or an odd number of b's.

$$b*(ab*ab*)* \mid a*ba*(ba*ba*)*$$